

Characteristics of Nerve Injury, Regeneration, and Vocal Fold Movement After Recurrent Laryngeal Nerve Electrocautery Injury[☆]

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SUMMARY: Objective. To investigate the temporal patterns of regeneration and functional recovery following a unilateral recurrent laryngeal nerve (URLN) electrocautery injury.

Methods. Unilateral recurrent laryngeal nerve (RLN) injury rabbit models (n = 12) were established. At 2, 4, 8, and 12 weeks post-injury, RLN and neuromuscular junction regeneration were evaluated via transmission electron microscopy and immunofluorescence. Thyroarytenoid muscle histopathology was assessed by hematoxylin and eosin staining, and vocal fold (VF) movement was quantified via laryngoscopy. Correlation between neuroelectrophysiological changes and VF motion was analyzed in eight patients with RLN electrocautery injuries.

Results. At 2 weeks, myelin sheaths and synaptic membranes showed extensive disintegration. The single muscle fiber diameter (SMFD) decreased to 56.16%, the muscle fiber area ratio (MFAR) decreased to 40.35%, and the motion range ratio (MRR) decreased to 12.2%. At 4 weeks, the myelin lamellae continued to disintegrate with a small amount of regeneration. Regeneration of the presynaptic membrane (more than the postsynaptic membrane) began. MFAR was 46.86%, and MRR was 77.9%. At 8 weeks, the density of myelinated axon (DMA) peaked, but the axons and myelin sheath remained thin and immature. Postsynaptic membrane regeneration was significant, with partial overlap by the presynaptic membrane. MFAR rose to 58.59%, and MRR recovered to 96.9%. At 12 weeks, myelin sheaths thickened and matured despite a drop in DMA. Synaptic membrane number and morphology were normalized. SMFD and MFAR neared normal levels at 94.52% and 91.57%, respectively, and MRR recovered fully. 77.8% of patients exhibited partial or complete VF movement recovery.

Conclusion. Despite causing initial injury, electrocautery may still permit sufficient axonal regeneration and subsequent muscle reinnervation to enable partial or complete recovery of VF movement.

Keywords: Recurrent laryngeal nerve–Electrocautery–Vocal fold paralysis–Transmission electron microscopy–Immunofluorescence–Vocal fold movement.

INTRODUCTION

Iatrogenic laryngeal nerve injuries, which frequently occur during thyroid surgery,^{1–5} can lead to vocal fold (VF) paralysis/paresis, dysphagia, dysphonia, and/or dyspnea, severely impacting patients' quality of life.

Thermal injury to the RLN is common in clinical practice, with prognosis varying considerably depending on factors such as the type of cautery instrument and the location and severity of nerve damage. Yang X⁶ demonstrated that there were no significant functional or histological changes induced by the electrotome at a vertical distance of 5 mm or 3 mm from the RLN. The electrotome (15 kW) caused more

serious thermal damage to thyroid tissues than that caused by either the Harmonic scalpel or LigaSure. Best CAE⁷ further demonstrated that the rates of postoperative RLN injury did not differ significantly between the Harmonic Scalpel and LigaSure. Zohar Y⁸ demonstrated that Unipolar coagulation caused significant Schwann cell injury, whereas bipolar coagulation preserved nerve architecture, especially in neurovascular-rich areas. According to a meta-analysis by Tseimakh AE,⁹ both precise ultrasonic dissection and electric cauterization showed similar effectiveness in preventing recurrent laryngeal nerve injury. Studies by Dionigi G¹⁰ and Liu N¹¹ both identified thermal injury as a cause of recurrent laryngeal nerve injury during thyroid surgery. However, the reported incidence differed, with Dionigi G finding a higher rate (17%) compared to Liu N (7%). Dionigi G further specified that, among these thermal injuries, 72% were temporary and 28% were permanent vocal cord paralyses.

Currently, research on the mechanisms of nerve injury and regeneration following electrocautery is limited, and its effects on VF movement remain unclear. In this study, an animal model was established to simulate intraoperative URLN electro-coagulation injury. Time-dependent changes in the nerve, neuromuscular junctions (NMJs), and muscle were systematically observed, while VF recovery was monitored, aiming to provide a theoretical basis for individualized clinical management of RLN injury.

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MATERIALS AND METHODS

Establishment of URLN electrocautery injury models

Fourteen male New Zealand white rabbits (3 months old, $\approx 3 \pm 0.2$ kg) were selected for this study, provided by Hangzhou Fuyang Hongfeng Rabbit Breeding Farm (license number: SCXK (Zhe) 2020-0004). Twelve rabbits were used to establish the URLN electrocautery injury model groups, while two rabbits served as controls.

Rabbits were anesthetized using a combination of intramuscular injection of Su-Mian-Xin (0.1 ml/kg) and slow intravenous injection of 3% pentobarbital (0.3–0.5 ml/kg) via the marginal ear vein. Preoperative laryngoscopy was performed to confirm normal bilateral VF movement. At the level of the fourth tracheal ring, the nerve was easily identified and dissected over a length of 1–1.5 cm. The right RLN was directly cauterized using monopolar coagulation at 1 W power for 1 second, repeated 3 times (Figure 1). Ethics Number: xzmzsyky Ethics Review No. 2022-125.

Transmission electron microscopy (TEM)

For TEM analysis, tissue segments (≈ 1 cm) encompassing the RLN injury site were harvested at 2, 4, 8, and 12 weeks post-injury. Following fixation and processing into 1-mm³ pieces, ultrathin sections (60 nm) were prepared. These sections were stained and visualized under a transmission electron microscope (HT7800/HT7700, HITACHI, Japan). Morphometric analysis was performed using Image-J software at $\times 20 \mu\text{m}$ magnification to measure myelinated nerve fiber and axon diameters, from which cross-sectional areas were derived. Subsequently, key regenerative metrics—myelinated axon density (MAD; axons/unit area) and the N-ratio (myelinated axon area/total nerve trunk area)—were quantified to assess nerve injury and regeneration.

Immunofluorescence staining was used to analyze degradation and regeneration changes at the NMJs of the TA muscle

To assess changes at the Thyroarytenoid (TA) muscle NMJs, immunofluorescence (IF) staining was conducted. Following decalcification of the intact rabbit larynges,

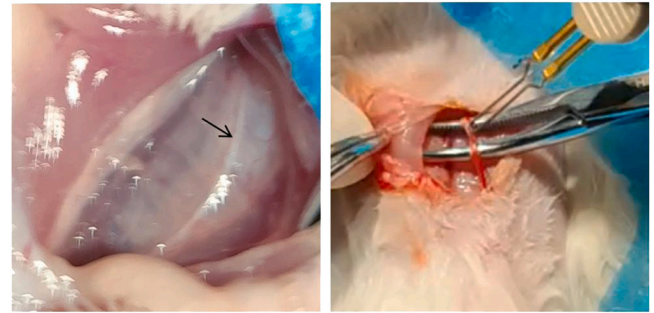


FIGURE 1. URLN electro-coagulation injury model, the black arrow indicates RLN.

tissues were paraffin-embedded and sectioned. Staining was performed using the following antibodies: anti-NF-H (Novus, NBP1-97726, 1:500) to label neurofilaments (yellow), anti-SynL2 (Novus, NBP1-77362, 1:500) to label presynaptic membranes (red), and anti-acetylcholine receptor (AChR) (Genetex, GTX22804-100ul, 1:5000) to label postsynaptic membranes (green). *ImageJ* software was then employed to separately quantify the numbers of neurofilaments, presynaptic structures, and postsynaptic structures, thereby evaluating NMJ injury and regeneration dynamics across the postoperative time course.

Evaluation of VF motion (Figure 2)

Transoral laryngoscopy combined with a flexible endoscopic imaging system was used to record bilateral VF motion in live rabbit models during quiet breathing at 2, 4, 8, and 12 weeks postoperatively. Video recordings (24 frames per second, 1 minute duration) were obtained, from which stable and clear 10-second segments were selected for analysis.

The range of VF motion (MR) was quantified by measuring the angular displacement from the open to closed phase. $MR_{\text{affected}} = \beta_1 - \beta$, $MR_{\text{normal}} = \alpha_1 - \alpha$ (as illustrated in Figure 2c). To control for anesthetic depth, individual differences, and examination differences, the contralateral healthy VF in each rabbit served as the internal control. Recovery of VF motion was evaluated using the Motion Range Ratio ($MRR = MR_{\text{affected}} / MR_{\text{normal}}$).

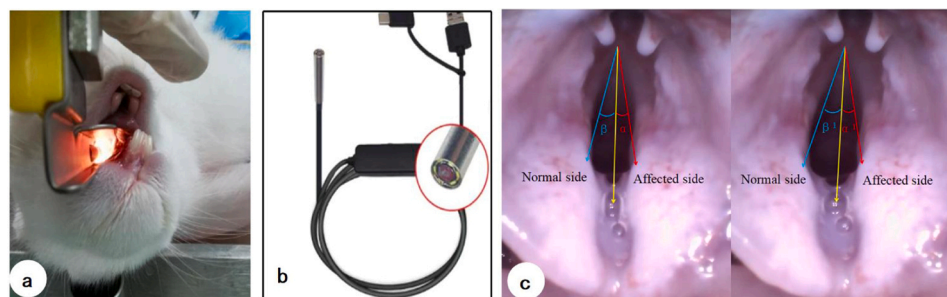


FIGURE 2. (a) Transoral laryngoscopy. (b) Flexible endoscopic imaging system. (c) Schematic diagram for quantifying VF motion, showing the angular change during the open-close cycle. The left panel represents the expiratory adduction phase, and the right panel represents the inspiratory abduction phase. VF, vocal fold.

Clinical data

Eight patients who sustained RLN electrocautery injuries and had complete medical records were enrolled in the study. Data on gender, age, laryngoscopy findings, and laryngeal electromyography (LEMG) were collected. LEMG (Nicolet EDX) was used to assess spontaneous potentials (SPs), motor unit action potentials (MUAPs), and recruitment patterns of the cricothyroid, TA, and posterior cricoarytenoid muscles during different physiological states, including rest, respiration, and phonation. Neurogenic injury was characterized by findings such as SPs, Long-duration MUAPs, or reduced recruitment (RR).

Statistical analysis

Statistical analysis was performed using SPSS (version 13.0). The Kruskal-Wallis H test was applied to compare the following parameters across different time points: MAD, N-ratios, NF-H, SypL2, AChR, single muscle fiber diameter (SMFD), and muscle fiber area ratio (MFAR). For MRR, One-way analysis of variance (ANOVA) was used, followed by Bonferroni-corrected *post*

hoc tests for multiple comparisons. A *P* value of less than 0.05 was considered statistically significant.

RESULT

TEM

Morphological observations (Figure 3): At 2 weeks, a large number of myelinated nerve fibers were damaged, with myelin sheaths disappearing, and a small number of partially damaged myelin sheaths were observed, which appeared "onion bulb-like". At 4 weeks, the number of regenerating myelinated fibers increased, although a few damaged myelinated fibers were still present. At 8 weeks, the quantity of regenerating myelinated fibers increased significantly, characterized by thin axons and thin myelin sheaths, and the MAD reached its peak. At 12 weeks, MAD decreased, but the myelin sheath thickness and the area of myelinated axons increased.

Morphometric Analysis (Figure 4): Myelin sheath thickness increased continuously from 4 weeks, with a statistically significant difference between 4 and 12 weeks.

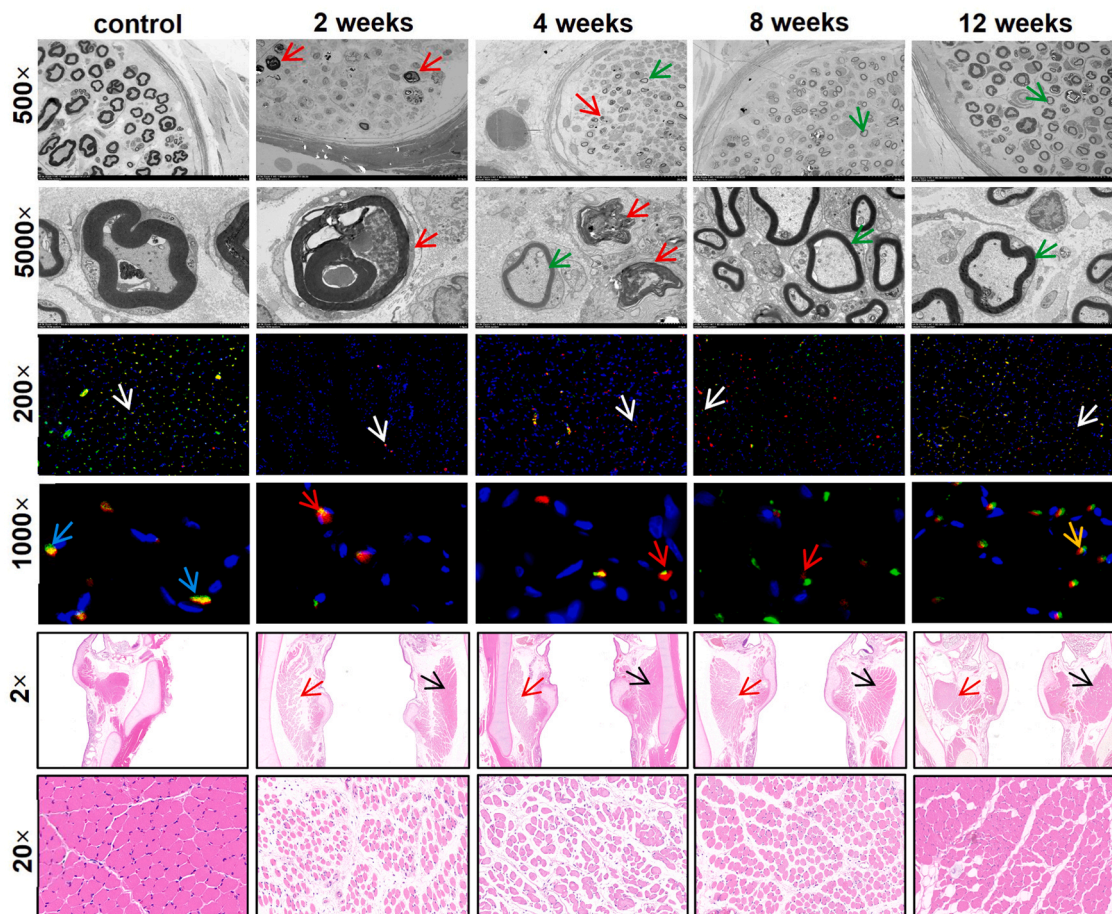


FIGURE 3. Electron micrographs of the RLN at different postoperative intervals (top two rows) show damaged (red arrows) and regenerated (green arrows) myelin sheaths. IF images of NMJs in the TA muscle (middle two rows) are displayed, with the fourth row showing a close-up of the region marked by the white arrow in the third row. NMJs are marked as normal (blue arrows), injured (red arrows), or regenerated (orange arrows). HE images demonstrating changes in the TA muscle (bottom two rows). The red and black arrows indicate the affected and the normal TA muscle, respectively. HE, hematoxylin and eosin; IF, immunofluorescence; TA, thyroarytenoid.

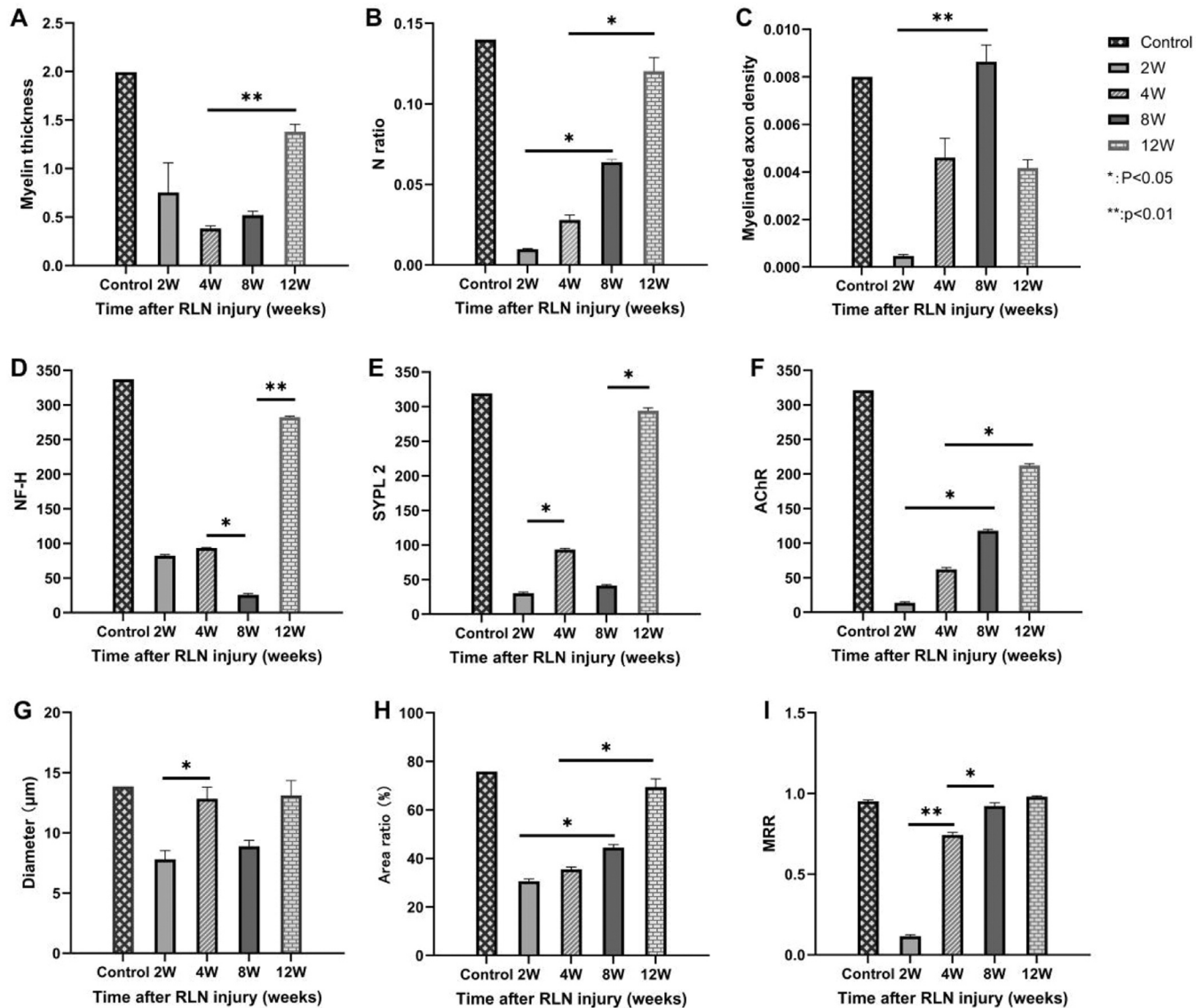


FIGURE 4. Comparison of myelin sheath thickness, N-ratio, MAD, NF-H, SynL2, AChR, SMFD, MFAR, and MRR at different time points postoperatively in a rabbit model. MAD, myelinated axon density; SMFD, single muscle fiber diameter; MFAR, muscle fiber area ratio; MRR, motion range ratio.

MAD increased from 2 weeks, peaking at 8 weeks, and declining by 12 weeks. The difference between 2 and 8 weeks was significant. The N ratio rose continuously from 2 weeks, reaching its highest at 12 weeks, with significant differences between 2 and 8 weeks and between 4 and 12 weeks. No statistically significant differences in the G ratio were observed across time points.

Immunofluorescence (Figure 3)

At 2 weeks, synaptic nerve endings and the pre and post-synaptic membranes were extensively degraded and nearly absent. NF-H decreased to 24.03%, SynL2 to 9.09%, and AChR to 4.36% of control levels. At 4 weeks, these structures gradually regenerated. Significant differences were observed for SynL2 between 2 and 4 weeks, and for AChR between 2 and 8 weeks (Figure 4). At 8 weeks, partial

overlap of the membranes was noted. At 12 weeks, NF-H recovered to 83.68%, SynL2 to 92.16%, and AChR to 66.98% of control levels. The overlap between the pre and postsynaptic membranes increased, and their morphology largely resembled that of the control group.

HE (Figure 3)

At 2 weeks, the SMFD decreased to 56.16% of control values. It then increased to 63.95% at 8 weeks and further to 94.52% at 12 weeks. The MFAR decreased to 40.35% at 2 weeks. Subsequently, it increased steadily from 2 to 12 weeks, reaching 91.57% at 12 weeks. Statistically significant differences in SMFD were observed between 2 and 4 weeks. For MFAR, significant differences were found between 2 and 8 weeks and between 4 and 12 weeks (Figure 4).

Vocal fold motion

All cases except one showed recovery beginning at 4 weeks. One case exhibited synkinesis at 8 weeks. MRR showed no statistically significant differences among preoperative, 8 weeks and 12 weeks. By week 8, the range of VF motion had largely recovered, but the movement remained asynchronous. Movement synchrony improved by week 12. Statistically significant differences in MRR were observed between 2 and 4 weeks, as well as between 4 and 8 weeks (Figure 4).

Clinical data analysis

A 3-month follow-up study was conducted on eight patients (nine VFs, including one bilateral case) who sustained RLN electrocautery injury during thyroid surgery (Table 1). The cohort consisted of seven females and one male, with a mean age of 37.5 ± 6.05 years. Injuries in three

patients (four VFs) occurred during open surgery with intraoperative neuromonitoring (IONM) using ultrasonic scalpels or bipolar electrocautery (Figure 5). The remaining five patients sustained injuries during thyroid ablation procedures. All cases were confirmed by LEMG, which revealed SPs in seven VFs and RR in all nine VFs. VF motion partially or fully recovered in 7/9 VFs (77.8%). However, 2/9 VFs (22.2%) showed no recovery after 3-month follow-up. The earliest recovery was observed at 4 weeks post-op, with motion amplitude largely restored by 8 weeks. Nonetheless, asynchronous movement between the bilateral VFs persisted.

DISCUSSION

Thermal nerve injury is a common iatrogenic complication caused by instruments such as unipolar coagulation,

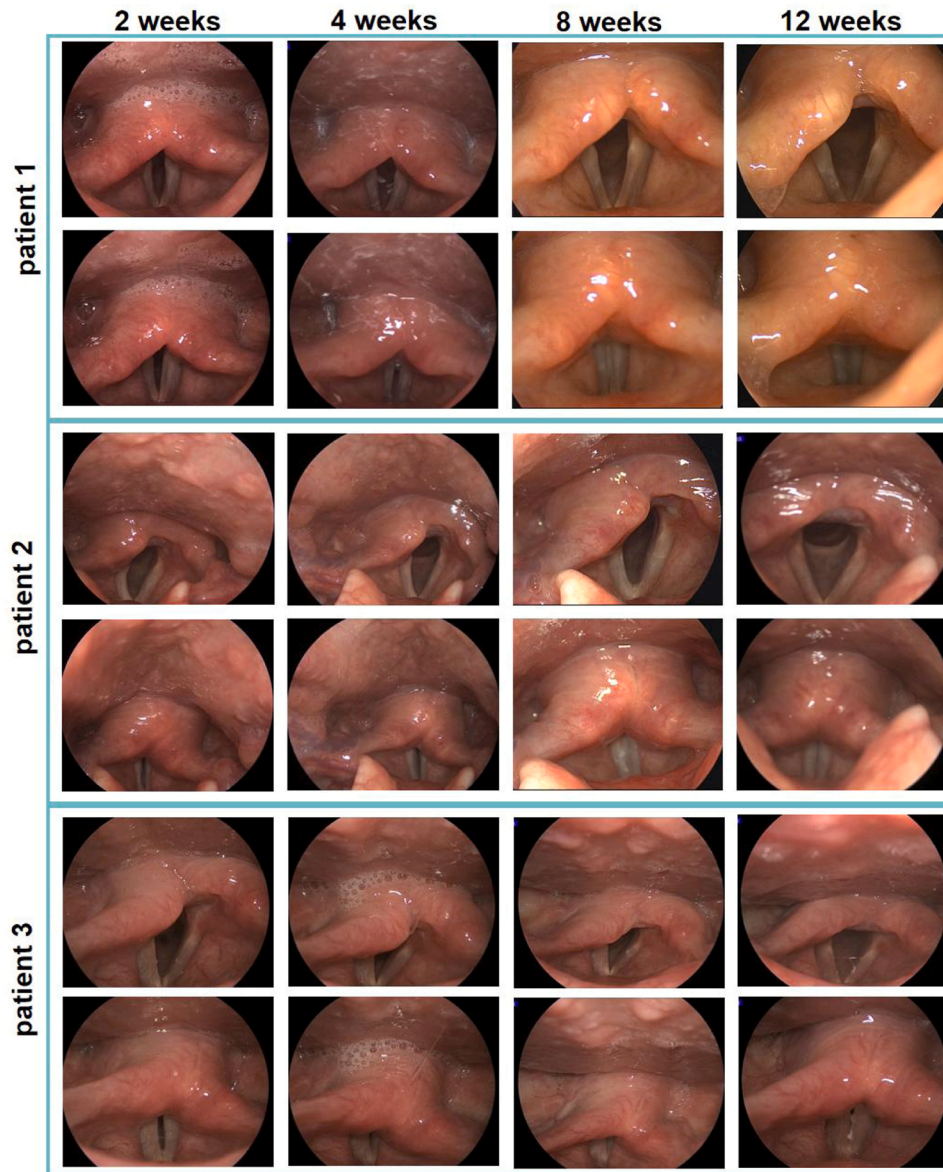


FIGURE 5. Sequential assessment of VF movement recovery in three patients (patient 1 with bilateral injury). VF, vocal fold.

bipolar coagulation, electrotome, ablation electrode, and ultrasonic scalpels.^{11,12} Such injury disrupts connections to target organs, leading to muscle atrophy and motor end-plate degeneration—changes that can be partially reversed if reinnervation occurs. The injury and repair processes vary depending on the etiology,^{10,13} which in turn influences functional outcomes. For example, the regeneration of motor nerve fibers proceeds at a slower rate after transection and repair (approximately 1–2 mm/day) compared to a crush injury, where the basal lamina tubes remain intact, allowing for a faster regeneration rate of 3–4 mm/day.¹⁴ Currently, research on the mechanisms of nerve regeneration after electrocautery injury to the RLN remains limited, and its impact on VF motion is still unclear.

Morphological observation is the first step in assessing nerve injury and regeneration. In this study, TEM of the RLN revealed extensive destruction and loss of myelinated axons at 2 weeks, indicating severe injury. Signs of regeneration emerged at 4 weeks, characterized by the presence of myelinated axons with relatively thin axons and thin myelin sheaths, suggesting a limited regeneration capacity. Regeneration peaked at 8 weeks, with MAD reaching its maximum—close to normal values—despite relatively low maturity. By 12 weeks, the myelin sheath thickness of the newly formed axons had increased significantly, and the maturity of the regenerated nerve improved, but MAD decreased compared to 8 weeks. MAD showed an initial increase followed by a decline, which is related to massive early axonal regeneration and later degeneration of mismatched axons. MAD differed significantly between 2 and 8 weeks ($P < 0.05$), and the thickness of the myelin sheath differed significantly between 4 and 12 weeks. These findings indicate that the early phase of nerve regeneration primarily involves a quantitative increase in axons, while the later phase focuses on qualitative maturation.

Previous studies have found that nerve injury leads to muscle atrophy and degeneration, characterized by decreased muscle cell volume and connective tissue proliferation. Muscle healing proceeds synchronously with nerve repair. If nerve repair is delayed, muscle atrophy and loss of physiological function will occur.¹⁵ This study found that at 2 weeks, the SMFD decreased to 56.16% and the MFAR decreased to 40.35%, indicating severe atrophy in the early stage. From 4 to 12 weeks, a trend of continuous recovery was observed. However, at 8 weeks, the SMFD showed a slight decrease to 63.95%, which may reflect fluctuations during the regeneration process. By 12 weeks, the SMFD had increased to 94.52%, and the MFAR recovered to 91.57%. These results demonstrate that nerve regeneration and successful reinnervation can occur after electrocautery injury.

The study also observed that at 2 weeks, the pre and postsynaptic membranes of the NMJ were extensively degraded. This timing coincided with the myelin sheath degeneration observed via TEM. From 2 to 4 weeks, the levels of NF-H, SynL2, and AChR all increased. However,

at 8 weeks, the levels of NF-H and SynL2 decreased, while AChR continued to rise. Regenerated pre and postsynaptic membranes showed partial overlap. By 12 weeks, the levels of NF-H, SynL2, and AChR were all increased significantly, the pre and postsynaptic membranes were fully aligned, and the number and morphology of NMJs resembled those on the healthy side. A notable finding was that the levels of NF-H, SynL2, and SMFD were all lower at 8 weeks than at 4 weeks. This decrease may be related to contact inhibition, a process in which regenerating axons, upon reaching muscle targets, induce the pruning of supernumerary branches.¹⁶

In this animal model study, VF motion partially or fully recovered in most cases, although synkinesis was observed in one instance. MRR was used to quantify changes in VF motion. No statistically significant differences in MRR were observed among the preoperative, 8-week, and 12-week postoperatively, indicating that the movement amplitude of the affected VF stabilized at 8 weeks. However, laryngoscopic observation revealed asynchrony in bilateral VF movement, with synchrony improving by week 12. Similar asynchronous movement has been reported in studies of RLN transection injuries,¹⁷ suggesting that the synchrony of bilateral VF movements may serve as an indicator for assessing nerve injury. Regarding recovery rates, a clinical study by Dionigi et al¹⁰ reported temporary and permanent VF paralysis rates of 72% and 28%, respectively, following thyroid ablation. In contrast, the recovery rates observed in our study were 77.8% (partial/full recovery) and 22.2% (no recovery).

Huang et al¹⁸ reported that the degree of nerve injury depends on ablation power and proximity to the nerve. In the present animal study, direct monopolar electrocautery was applied to the nerve (1 W for 1 second, repeated three times). Under these specific parameters, postoperative VF motion recovered favorably in all cases. These findings suggest that controlled, low-power electrocautery may not preclude functional recovery. However, further studies with larger cohorts are needed to substantiate these observations. Among the three patients who underwent IONM, VF motion also recovered well. IONM may facilitate the timely detection and prevention of further nerve injury, which may help mitigate the severity of damage.

LIMITATIONS

This study has several limitations. Firstly, as a pilot investigation, it was constrained by a modest sample size, necessitating larger future studies for validation. Secondly, the experiment was limited to a single power setting (1 W); thus, future research should incorporate variables like injury location, power levels, and device types to fully elucidate the mechanism of nerve damage. Third, a limitation of this study is that it did not assess how local micro-environmental changes (eg, collagen deposition and scar formation) influence nerve regeneration. Future work should incorporate evaluations, such as Masson's

TABLE 1.
Clinical Data of Patients with RLN Electrocautery Injury during Thyroid Surgery

Patient	Sex	Age	Instrument	Side	LEMG	VF motion recovery
1	F	32y	Harmonic scalpel	R	TA: SPs, RR PCA: SPs, Rare MUAPs	Partial
				L	TA: SPs, RR PCA: SPs, Polyphasic wave	Complete
2	F	40y	Bipolar coagulation	R	TA: RR, Rare MUAPs	Complete
3	F	33y	Harmonic scalpel	R	PCA: SP, RR, Rare MUAPs	Partial
4	F	34y	Ablation electrode	R	TA: RR	Complete
5	M	30y	Ablation electrode	R	TA: SPs, Rare MUAPs	Partial
6	F	42y	Ablation electrode	L	TA: SPs, RR PCA: SPs, Rare MUAPs	Partial
				R	TA: SPs, RR, Long-duration MUAPs	No
7	F	47y	Ablation electrode	R	TA: SPs, RR, Long-duration MUAPs	No
8	F	42y	Ablation electrode	L	TA: SPs, Polyphasic wave, RR PCA: Long-duration MUAPs, RR	No

Abbreviations: LEMG, laryngeal electromyography; MUAPs, motor unit action potentials; PCA, posterior cricoarytenoid; RR, reduced recruitment; SPs, spontaneous potentials; TA, thyroarytenoid.

trichrome staining, to quantify fibrosis and elucidate its role in the regenerative process.

CONCLUSION

Despite causing initial injury, RLN electrocautery may still permit sufficient axonal regeneration and subsequent muscle reinnervation to enable partial or complete recovery of VF movement.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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